

Project – Uber Trip Analysis

Introduction:

This Power BI implementation project focuses on analyzing Uber trip data to help stakeholders make data-driven decisions related to booking trends, revenue generation, trip efficiency, and operational optimization. The dashboards built for this project provide key performance indicators (KPIs), location and vehicle insights, time-based trends, and drill-through capabilities for in-depth exploration. Leveraging Power BI's dynamic capabilities, this solution empowers users to monitor business metrics, identify peak periods, optimize pricing models, and improve overall customer experience.

Steps:

1. Imported Data

- Loaded the following Excel sheets:
 - Uber Trip Details
 - Location Table
- Verified row counts match Excel file for data integrity.

2. Checked the Data Quality

- Opened Power Query Editor.
- Enabled *Column Quality*, *Column Distribution*, *Column Profile*.
- Ensured:
 - 100% Valid values
 - 0% Errors

3. Relationship Setup

- [Trip Details[PULocationID]] <— Location Table[LocationID]] (Active)
- [Trip Details[DOLocationID]] <— Location Table[LocationID]] (Inactive)
- One-to-Many, Single Direction relationships.

4. Created Calendar Table

- Extracted date and time from **Pickup Date**.
- Created a calculated table using DAX:
Calendar Table = `CALENDAR(MIN('Trip Details'[Pickup Date]), MAX('Trip Details'[Pickup Date]))`
- Created relationships with pickup_date.
[Trip Details[Pickup Date]] <— Calendar Table[Date]]

5. Created following KPI Measures:

- Total Bookings = `COUNT('Trip Details'[Trip ID])`
- Total Booking Value = `SUM('Trip Details'[fare_amount]) + SUM('Trip Details'[Surge Fee])`
- Avg Booking Value = `DIVIDE(SUM('Trip Details'[fare_amount]) + SUM('Trip Details'[Surge Fee]), COUNT('Trip Details'[Trip ID]), BLANK())`
- Total Trip Distance = `VAR TotalMiles = SUM('Trip Details'[trip_distance]) / 1000 RETURN CONCATENATE(FORMAT(TotalMiles, "0"), "K miles")`
- Total Trip Distance Measure = `SUM('Trip Details'[trip_distance])`
- Avg Trip Distance = `VAR TotalMiles = ROUND(AVERAGE('Trip Details'[trip_distance]), 0) RETURN CONCATENATE(TotalMiles, " miles")`
- Avg Trip Time = `VAR AvgMinutes = AVERAGEX('Trip Details', DATEDIFF('Trip Details'[Pickup Time], 'Trip Details'[Drop Off Time], MINUTE)) RETURN CONCATENATE(FORMAT(AvgMinutes, "0"), " min")`

6. Dashboard: Overview Analysis

- Added Card (new) to show:
 - Total Bookings
 - Total Booking Value
 - Avg Booking Amount
 - Total Trip Distance
 - Avg Trip Distance
 - Avg Trip Time
- Formatted the card contents

7. Created Dynamic Measure Selector

- Created Dynamic Measure:
 Dynamic Measure = {
 ("Total Bookings", NAMEOF("Trip Details'[Total Bookings]"), 0),
 ("Total Booking Value", NAMEOF("Trip Details'[Total Booking Value]"), 1),
 ("Total Trip Distance", NAMEOF("Trip Details'[Total Trip Distance Measure]"), 2)
 }
- Set Title for Dynamic Measure:
 Dynamic Title =
 IF(
 'Dynamic Measure'[Dynamic Measure Order] = 0,
 "Total Bookings",
 IF(
 'Dynamic Measure'[Dynamic Measure Order] = 1,
 "Total Booking Value",
 IF(
 'Dynamic Measure'[Dynamic Measure Order] = 2,
 "Total Trip Distance",
 "Other"
)
)
)
- Format the Dynamic Measure to show as slicer and single value

8. Donut Charts

- Created donut chart for Payment Type as Legend with values of Dynamic Measure.
- Created another donut chart for Trip Type (Day/Night) with below new column as Legend and the values of Dynamic Measure.
 Trip Type (Day/Night) =
 VAR HourOfDay = HOUR("Trip Details'[Pickup Time]")
 RETURN
 IF(HourOfDay >= 17 || HourOfDay < 6, "Night Trip", "Day Trip")

9. Vehicle Type Grid (Matrix):

- Added a Matrix by putting Vehicle in Rows and followings as Values:
 - Total Bookings
 - Total Booking Value
 - Avg Booking Amount
 - Total Trip Distance
- Added a title with Vehicle Type Title = "Vehicle Type Analysis" Measure
- Added Data Bar for Total Bookings by using conditional formatting

10. Area Chart

- Added an Area Chart by putting Day from Date Hierarchy of Calendar table in X-axes and Total Booking in Y-axes
- Done necessary formatting

11. Created following Measures Location Analysis:

- Most Frequent Dropoff Point =
 VAR DropOffCounts =
 ADDCOLUMNS(
 SUMMARIZE("Trip Details", 'Location Table'[Location]),
 "DropOffCount",
 CALCULATE(
 COUNT("Trip Details'[Trip ID]"),
 USERELATIONSHIP("Trip Details'[DOLocationID]", 'Location Table'[LocationID])
)
)
 VAR RankedDropoffs =
 ADDCOLUMNS(DropOffCounts, "Rank", RANKX(DropOffCounts, [DropOffCount],, DESC, DENSE))
 VAR TopDropoff = FILTER(RankedDropoffs, [Rank] = 1)
 RETURN
 CONCATENATEX(TopDropoff, 'Location Table'[Location], ", ")
- Most Frequent Pickup Point =
 VAR MostPickup =
 TOPN(1,

```

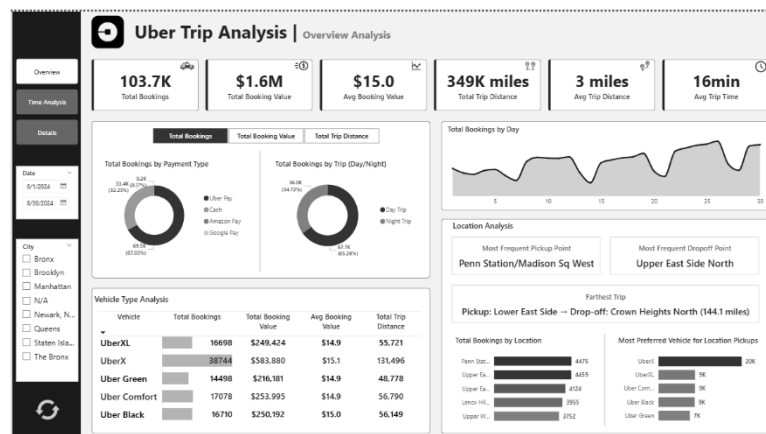
SUMMARIZE(
    'Trip Details',
    'Location Table'[Location],
    "PickupCount", COUNT('Trip Details'[Trip ID])
),
[PickupCount], DESC
)
RETURN
CONCATENATEX(MostPickup, 'Location Table'[Location], ", ")
o Farthest Trip =
VAR MaxDistance = MAX('Trip Details'[trip_distance])
VAR PickupLocation =
    LOOKUPVALUE(
        'Location Table'[Location],
        'Location Table'[LocationID],
        CALCULATE(
            SELECTEDVALUE('Trip Details'[PULocationID]),
            'Trip Details'[trip_distance] = MaxDistance
        )
    )
VAR DropoffLocation =
    LOOKUPVALUE(
        'Location Table'[Location],
        'Location Table'[LocationID],
        CALCULATE(
            SELECTEDVALUE('Trip Details'[DOLocationID]),
            'Trip Details'[trip_distance] = MaxDistance
        )
    )
RETURN
    "Pickup: " & PickupLocation & " → Drop-off: " & DropoffLocation & " (" & FORMAT(MaxDistance, "0.0") &
    " miles)"

```

- Added a Card with the Data of Most Frequent Pickup Point and Most Frequent Dropoff Point
- Done necessary formatting
- Added another Card with the Data of Furthest Trip
- Done necessary formatting
- Added a Stacked Bar Chart to show Total Bookings by Location (Top 5) by putting Location in Y-axes and Total Booking in X-axes
- Done necessary formatting
- Added another Stacked Bar Chart to show Most Preferred Vehicle for Location Pickup by putting Vehicle in Y-axes and Count of PULocationID in X-axes
- Done necessary formatting
- Added a title with Locaton Type Title = "Location Analysis" Measure

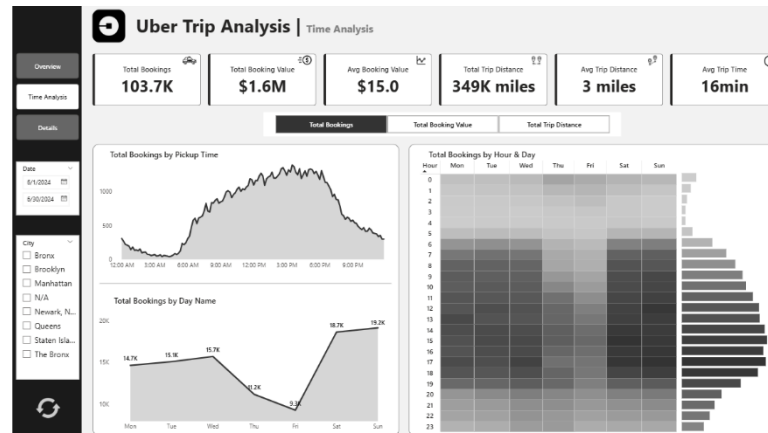
12. UX Features:

- Added Slicers with Date and City
- Edited interactions



13. Dashboard: Time Analysis

- Duplicated the Overview Analysis Dashboard
- Deleted irrelevant contents
- Added an Area Chart to show Pickup Time (10-Minute Intervals) by creating a new group form Pickup Time with 10 bins.
- Added a Line Chart to show By Day Name by adding new column Day and Day num in Calendar Table
- Added a Matrix By Hour and Time by adding Pickup Hour = `HOUR('Trip Details'[Pickup Time])` in Rows and Days in Columns.
- Added a Stacked Bar Chart by adding Pickup Hour Measure in Y-axis and Dynamic Measures in X-axis



14. Dashboard: Details

- Duplicated the Overview Analysis Dashboard
- Created Grid Table include key fields (Trip ID, Pickup Date, Vehicle, Payment Type, PU/DO Location, etc.)
- Added Bookmarks
- Added Page Navigators
- Added a Reset icon with action Clear all slicers.

Trip ID	Pickup Date	Vehicle	Payment type	Passengers Count	Total Trip Distance	Total Booking Value	Pickup Location	Dropoff Location	Total B.
1	Saturday, June 01, 2024	UberX	Uber Pay	1	5.60	\$21.5	East Village	Sunnyside	
2	Saturday, June 01, 2024	Uber Black	Cash	1	1.72	\$8.0	Lincoln Square East	Penn Station/Madison Sq West	
3	Saturday, June 01, 2024	Uber Black	Cash	1	3.41	\$13.0	Sutton Place/Turtle Bay North	Upper West Side North	
4	Saturday, June 01, 2024	UberX	Cash	1	1.81	\$9.0	Prospect Lefferts Gardens	Brownsville	
5	Saturday, June 01, 2024	Uber Black	Cash	1	1.89	\$8.0	Garment District	Kips Bay	
6	Saturday, June 01, 2024	UberX	Cash	6	3.29	\$14.0	Central Harlem	Lincoln Square West	
7	Saturday, June 01, 2024	UberX	Cash	2	2.05	\$8.5	Lincoln Square East	Lenox Hill West	
8	Saturday, June 01, 2024	Uber Comfort	Cash	2	3.54	\$12.5	Clinton East	East Harlem South	
9	Saturday, June 01, 2024	Uber Green	Cash	1	1.10	\$5.5	Clinton East	West Chelsea/Hudson Yards	
10	Saturday, June 01, 2024	UberX	Uber Pay	2	1.90	\$16.5	Lenox Hill East	Times Sq/Theatre District	
11	Saturday, June 01, 2024	UberXL	Uber Pay	1	6.66	\$28.5	Clinton East	Central Harlem North	
12	Saturday, June 01, 2024	UberX	Uber Pay	2	13.12	\$36.5	Kips Bay	Parkchester	
13	Saturday, June 01, 2024	Uber Green	Cash	1	12.59	\$37.0	JFK Airport	Prospect Heights	
14	Saturday, June 01, 2024	UberXL	Cash	1	1.10	\$5.5	Clinton East	Lincoln Square East	
15	Saturday, June 01, 2024	UberX	Cash	1	6.00	\$27.0	Mountingdale Heights	Lower East Side	
16	Saturday, June 01, 2024	Uber Black	Uber Pay	1	2.12	\$10.4	Kips Bay	Lenox Hill East	
17	Saturday, June 01, 2024	UberXL	Uber Pay	1	3.00	\$13.0	Midtown North	East Village	
18	Saturday, June 01, 2024	Uber Comfort	Uber Pay	1	3.10	\$14.6	Greenwich Village South	Seaport	
19	Saturday, June 01, 2024	UberXL	Uber Pay	1	5.83	\$21.3	East Williamsburg	Murray Hill	
20	Saturday, June 01, 2024	Uber Green	Uber Pay	1	1.53	\$8.6	Clinton East	Union Sq	
21	Saturday, June 01, 2024	Uber Comfort	Cash	1	1.70	\$8.5	Fort Greene	Prospect Heights	
22	Saturday, June 01, 2024	UberXL	Uber Pay	1	5.80	\$25.5	Penn Station/Madison Sq West	Williamsburg (North Side)	
23	Saturday, June 01, 2024	Uber Pay	Uber Pay	1	2.10	\$10.5	Penn Station/Madison Sq West	Greenwich Village South	
24	Saturday, June 01, 2024	Uber Comfort	Uber Pay	1	4.19	\$21.0	Clinton East	Seaport	
25	Saturday, June 01, 2024	UberX	Cash	1	1.08	\$7.0	East Village	Gramercy	
26	Saturday, June 01, 2024	Uber Black	Cash	1	1.80	\$7.0	Clinton East	Meatpacking/West Village West	
27	Saturday, June 01, 2024	Uber Green	Cash	1	1.80				
Total					348,933.81	\$1,553,672.6			

Key Insights:

1. Booking & Revenue Overview

- **Total Bookings:** 103.7K trips, generating **\$1.6M** in revenue.
- **Avg Booking Value:** **\$15.0**, with **UberX** being the most booked (38,744 trips, **\$583,880** revenue).
- **Avg Trip Distance:** **3 miles**, with an **Avg Trip Time** of **16 minutes**.
- **Farthest Trip:** Lower East Side → Crown Heights North (144.1 miles).

2. Vehicle Type Analysis

- **Most Popular Vehicle: UberX** (37.4% of bookings), followed by **UberXL** (16.1%) and **Uber Comfort** (16.5%).
- **Revenue Contribution:**
 - **UberX** (\$583,880)
 - **Uber Black** (\$250,192)
 - **UberXL** (\$249,424)
- **Avg Fare Consistency:** Most vehicle types average **\$14.9–\$15.1** per trip.

3. Time & Demand Patterns

- **Peak Hours:** Likely **morning & evening rush hours** (data suggests spikes around **8 AM to 6 PM**).
- **Weekend Demand:** **Sundays** show high activity

4. Location & Route Insights

- **Top Pickup/Dropoff Points:**
 - **Pickup: Penn Station/Madison Sq West** (most frequent).
 - **Dropoff: Upper East Side North** (most frequent).
- **High-Demand Areas:**
 - **Penn Station** (4,475 trips)
 - **Upper East Side** (4,459 trips)
- **Preferred Vehicle by Location:**
 - **UberX** dominates in most areas (~20K pickups).
 - **UberXL, Comfort** and **Uber Black** are secondary choices (~9K each).

5. Actionable Takeaways

- **Optimize UberX Availability** (highest demand).
- **Increase Promotions for Off-Peak/Night Rides** (to balance demand).
- **Encourage Digital Payments** (reduce reliance on cash).
- **Focus on High-Traffic Zones** (Penn Station, Upper East Side).
- **Monitor Long-Distance Trips** (potential for premium pricing strategies).
- This data suggests **UberX is the backbone of operations**, with **peak demand during weekdays and key transit hubs**. Adjusting driver allocation and pricing in high-demand areas could further boost efficiency.

Conclusion:

The Uber Trip Analysis dashboards effectively transform raw trip data into actionable insights. Using Power BI's dynamic visualizations, interactive filters, and advanced DAX measures, this solution delivers comprehensive views of trip performance, booking behavior, and operational patterns. Stakeholders can now make informed decisions about pricing, fleet distribution, and customer engagement strategies based on real-time, data-driven findings. This project also lays a foundation for scaling with predictive analytics, automation, and deeper customer segmentation in the future.